

Transport-State Fusion for Fast Meyer–Bregman Decomposition

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Abstract

Efficient cartoon–texture decomposition requires more than inexpensive iterations: the information needed by the next coupled subproblem must survive the current one. We present an engineering realization of Meyer–Bregman decomposition organized around a six-field transport state. Two primal images and four accumulated gradient fields close a warm interleaved recurrence. Retained primal spectra, streamed reflected divergences, and a lower-triangular spectral solve eliminate redundant work without changing that recurrence. The construction arose from studying transport between related measurements in an earlier superresolution experiment; matrix-function approximation was subsequently used to advance the retained state, rather than to define it. We distinguish exact implementation improvements from approximate trajectory acceleration. A finite-flow polynomial on the complete state avoids the specific retained-carrier and contour-halo defects of an earlier scalar surrogate. Its numerical machinery is standard Krylov projection; the contribution is the state formulation, its fused realization, and the measured tradeoff on this decomposition. A September audit adds denser ordinary controls, late trajectory checkpoints, and inexpensive feasible primal–dual certificates. Agreement with a visually accepted 64-pass image is not equivalent to objective convergence. A current-state factor experiment improves finite-reference accuracy, but fixed factors have parameter-dependent counterexamples. We identify statistically determined, certified, low-overhead step selection as an open engineering requirement. Code, numerical oracles, timing samples, and negative controls accompany the paper.

Keywords: Meyer decomposition, bounded flux, Split Bregman, transport state, spectral fusion, finite flow, primal–dual certificate.

1 Introduction

Meyer’s model represents oscillation through the divergence of a bounded vector field, rather than by a frequency cutoff [1, 2]. Gilles and Osher used duality and Split Bregman to implement the associated alternating ROF subproblems [5, 4]. That model and its Bregman interpretation are the starting point of this work.

Our engineering question is what must remain available when one coupled subproblem becomes the next. The driving images change during the alternation. An estimate of the image alone does not necessarily retain the flux and projection information that makes the next update effective. Repeated cold solves discard useful information; extrapolation of stale subproblem updates can carry information in the wrong frame.

The development began with transform planning and a two-lattice superresolution experiment concerned with transport between related measurements. In the Meyer problem, this motivated retaining the coupled transport state and reducing repeated work around it. The resulting sequence was warm interleaving, six-field reduction, retained spectra, paired spectral solution, and only then finite-state advance. Transport here refers to the model’s bounded-flux structure and propagation of its coupled state. We do not claim a new optimal-transport distance or a mass-preserving image warp.

This distinction matters for attribution. The transport construction was the author’s motivating idea. Krylov projection is a recognized numerical tool used in its later implementation, not the claimed source of that idea. Conversely, a different route of discovery does not make established matrix-function algebra new.

The engineering contributions are:

1. A closed warm six-field recurrence and its exact spectral fusion, including source initialization and effective two-product output.
2. A diagnosis of a scalar shortcut’s failure and a finite-flow implementation on the retained state, with independent checks.
3. A measurement framework separating elapsed work, finite-trajectory agreement, and objective certificates, including negative results.

The September revision adds a current-state factor experiment and explicit requirements for dynamic factors. These are subsequent investigations, not retroactive explanations of the August measurements.

1.1 Relation to prior work

The Meyer model, Chambolle projection, and Split Bregman are established [1, 2, 3, 4]. Gilles–Osher’s Bregman implementation appeared as UCLA CAM Report 11-73 in 2011; its 2024

arXiv posting is later dissemination [5]. Fourier diagonalization, warm starts, and Krylov matrix-function actions are established techniques [7, 8]. Accelerating splitting states has close precedents in Anderson-accelerated Douglas–Rachford and ADMM [9, 10]. Those are relevant comparisons, although they do not identify the finite polynomial or exact realization here.

A changing inner driving image does not make all acceleration of a fully specified enlarged fixed-point map impossible. It does invalidate treating successive inner problems as one unchanged problem without justification. Historical experiments motivating this work rejected stale subproblem momentum. The present method reconstructs the current coupled update.

2 The retained transport state

Let $X = \mathbb{R}^{H \times W}$ with periodic forward gradient $D : X \rightarrow X^2$ and D^* its adjoint. We use $\text{div} = -D^*$ throughout. The discrete isotropic variation is

$$J(u) = \text{TV}(u) = \sum_x \|(Du)(x)\|_2. \quad (1)$$

Meyer’s discrete oscillation norm is

$$\|v\|_G = \inf_{v=D^*p} \|p\|_{\infty,2}, \quad \|p\|_{\infty,2} = \max_x \|p(x)\|_2. \quad (2)$$

The associated BV– G formulations and alternating projections are developed in [1, 2, 6]. We use the same [0, 255] intensity convention and fixed parameters $\lambda = .05, \mu = 40$.

The optimized reference retains two primal fields and four accumulated gradient fields:

$$z = (u, w, t_u^x, t_u^y, t_w^x, t_w^y). \quad (3)$$

Here $w = f - u - v$ is the finite-pass survivor. Consequently the public two-product decomposition is

$$v = f - u - w, \quad \bar{u} = f - v = u + w. \quad (4)$$

This convention matters in comparisons: raw u from a finite model is not the effective cartoon $f - v$.

Define the pointwise disk projection and reflected projection

$$\Pi_a(t) = \begin{cases} t, & |t| \leq a, \\ a, t/|t|, & |t| > a, \end{cases} \quad R_a = I - 2\Pi_a, \quad (5)$$

and the screened inverse $S(c, \eta) = (cI + \eta D^* D)^{-1}$. With

$$\begin{aligned} c_u &= \lambda, & \eta_u &= 2\lambda, & a_u &= \eta_u^{-1}, \\ c_w &= \mu^{-1}, & \eta_w &= 10\mu^{-1}, & a_w &= \eta_w^{-1}, \end{aligned} \quad (6)$$

Write $S_u = S(c_u, \eta_u)$ and $S_w = S(c_w, \eta_w)$. One exact warm fused pass $T_f : z \mapsto z^+$ is

$$u^+ = S(c_u, \eta_u)\{c_u(u + w) + \eta_u D^* R_{a_u}(t_u)\}, \quad (7)$$

$$w^+ = S(c_w, \eta_w)\{c_w(f - u^+) + \eta_w D^* R_{a_w}(t_w)\}, \quad (8)$$

$$t_u^+ = Du^+ + \Pi_{a_u}(t_u), \quad t_w^+ = Dw^+ + \Pi_{a_w}(t_w). \quad (9)$$

Proposition 2.1 (Lower-triangular spectral pass). *For periodic boundaries, equations (7)–(8) form a lower-triangular two-by-two system at every Fourier coefficient. Both reflected divergences may be transformed before either inverse transform, and the two primal spectra may be solved in one range traversal.*

Proof. $D^* D$ diagonalizes under the periodic DFT. If σ_u and σ_w are the two screened multipliers and $\hat{d}_i$ denotes the transform of $D^* R_{a_i}(t_i)$, then

$$\begin{aligned} \hat{u}^+ &= \sigma_u\{c_u(\hat{u} + \hat{w}) + \eta_u \hat{d}_u\}, \\ \hat{w}^+ &= \sigma_w\{c_w(\hat{f} - \hat{u}^+) + \eta_w \hat{d}_w\}. \end{aligned}$$

The second equation depends on the newly computed first variable and no reverse coupling remains. This is exactly the paired spectral loop in the C++ kernel. $\square$

2.1 Source initialization

The first pass is source driven:

$$\begin{aligned} u_1 &= S_u(c_u f), & w_1 &= S_w\{c_w(f - u_1)\}, \\ t_{u,1} &= Du_1, & t_{w,1} &= Dw_1. \end{aligned} \quad (10)$$

It is not obtained by inserting a zero state into (7). Every experimental pass count includes this initialization. The plus signs before $D^* R$ in the recurrence correspond to the code’s $-\eta \text{div } R$. Exact recomposition does not itself certify that the finite emitted texture lies in the prescribed G ball.

3 Exact implementation economy

The current driving images are $u + w$ and $f - u^+$. They change, but the fields needed to compute them and their Bregman corrections remain in the state. Warm interleaving takes one inner sweep per route and immediately updates the coupled state. It is a different finite schedule from converging both inner problems at every outer step.

The screened solve resolves global spatial linear coupling exactly for the selected boundary conditions. This can replace many local updates even when each local update is individually cheap. The retained state and global solve work together; neither a warm start nor a cheap local step alone explains the efficiency.

Retaining $\hat{f}, \hat{u}, \hat{w}$ avoids transforming driving images already known spectrally. Streaming the reflected divergence avoids materializing unneeded intermediate fields. Pairing the spectral solution shares traversal and synchronization work. An ordinary outer pass contains two forward and two inverse two-dimensional transforms, one pair per screened subproblem.

These are exact changes of representation and execution of the chosen recurrence. They must be distinguished from taking fewer nonlinear passes or approximating several passes by a jump. Persistent workers and reusable plans support the native implementation. The one-axis FACR route extends shape support, but the finite-flow tangent remains full spectral.

4 Diagnosis of the scalar shortcut

The rejected hard construction replaced the state recurrence by two scalar observations. Let $\mathcal{S}$ be its scalar separator, H its contraction, $P_K = I - H^K$, and

$$\begin{aligned} s_0 &= \mathcal{S}(f), & m_0 &= P_K(f - s_0), \\ r_1 &= f - m_0, & s_1 &= \mathcal{S}(r_1). \end{aligned} \quad (11)$$

Before its final G -capacity correction, the texture proposal is $m = P_K(f - s_1)$. Write $v_H = m + q_\mu$, where q_μ is the change caused by that final capacity route.

Proposition 4.1 (Common cause of halo and retained texture). *Let (u_*, v_*) be any exact two-product reference, $f = u_* + v_*$. The second hard observation and its final texture error satisfy*

$$r_1 = u_* + H^K v_* + P_K(s_0 - u_*), \quad (12)$$

$$v_H - v_* = P_K(u_* - s_1) - H^K v_* + q_\mu. \quad (13)$$

Proof. Since $P_K = I - H^K$,

$$\begin{aligned} r_1 &= f - P_K(f - s_0) = s_0 + H^K(f - s_0) \\ &= u_* + H^K v_* + P_K(s_0 - u_*), \end{aligned}$$

which proves (12). Similarly,

$$\begin{aligned} m - v_* &= P_K(f - s_1) - v_* \\ &= P_K(u_* - s_1) + (P_K - I)v_* \\ &= P_K(u_* - s_1) - H^K v_*. \end{aligned}$$

Adding $q_\mu = v_H - m$ proves (13). $\square$

The two visible failures are therefore one representation error. $H^K v_*$ is the uncompleted carrier tail; it remains in cartoon with the opposite sign in texture. If $e = u_* - s_1$ contains a step mismatch and $H^K e = h_K * e$ for a normalized smoothing kernel h_K , then $P_K e = e - h_K * e$ has the paired positive and negative contour response of a halo. It has zero mean, so a one-sided clamp cannot remove it faithfully. For a Fourier carrier with contraction gain $0 < \rho < 1$, $H^K v_* = \rho^K v_* \neq 0$ at every finite K .

Figure 1 renders the terms independently. Across six 256² controls, direct substitution into (12) and (13) leaves at most 1.42×10^{-13} in L_∞ . This is an identity check, not a fitted explanation.

The identity explains the observed defects of this construction. It is not an impossibility theorem for all scalar algorithms, nor proof that a different representation cannot produce a halo by another mechanism. The engineering response was to advance the actual state instead of repairing the output using a content-admission rule.

5 Finite flow of the complete state

After a short ordinary prefix, fix the current state z and define

$$r = T_f(z) - z, \quad A \in \partial T_f(z). \quad (14)$$

If T_f were affine on the visited chart, its exact finite displacement after m more passes would be

$$T_f^m(z) - z = p_m(A)r, \quad p_m(A) = I + A + \cdots + A^{m-1}. \quad (15)$$

This is the finite-trajectory target. The stationary Newton displacement $(I - A)^{-1}r$ was tested and rejected: it can overshoot a desired finite state at structural transitions.

The only nonlinear branch derivative comes from the disk projection already present in (5). For $n = t/|t|$,

$$D\Pi_a(t)h = \begin{cases} h, & |t| < a, \\ \frac{a}{|t|}(I - nn^T)h, & |t| > a, \end{cases} \quad (16)$$

and $DR_a(t)h = h - 2D\Pi_a(t)h$. At $|t| = a$ the implementation chooses the interior member of the Clarke derivative. Differentiating the fused triangle gives

$$\begin{aligned} \dot{u}^+ &= S_u\{c_u(\dot{u} + \dot{w}) + \eta_u D^* DR_{a_u}(t_u)\dot{t}_u\}, \\ \dot{w}^+ &= S_w\{-c_w\dot{u}^+ + \eta_w D^* DR_{a_w}(t_w)\dot{t}_w\}, \\ \dot{t}_u^+ &= D\dot{u}^+ + D\Pi_{a_u}(t_u)\dot{t}_u, \\ \dot{t}_w^+ &= D\dot{w}^+ + D\Pi_{a_w}(t_w)\dot{t}_w. \end{aligned} \quad (17)$$

No statistic of f enters (17); only the current optimization state and the original projection geometry do.

5.1 A standard small numerical representation

The implementation evaluates r , Ar , and A^2r . Let

$$\begin{aligned} B &= [r, Ar], & G &= B^T B, \\ K &= B^T A B, & \hat{A} &= G^{-1}K. \end{aligned} \quad (18)$$

Because Ar is the second basis vector, the first column of $\hat{A}$ is exactly $(0, 1)^T$. Only the projection of A^2r needs a two-by-two solve. The applied displacement is

$$\delta_m = B p_m(\hat{A}) e_1. \quad (19)$$

This is standard Krylov matrix-function machinery [7, 8], not a newly claimed algebraic principle. It is algebraically the depth-two Arnoldi polynomial without normalization or full-field modified Gram–Schmidt sweeps. A fixed 64-chunk reduction computes the five required inner products in a thread-count-independent summation order.

5.2 What error remains

Write the semismooth expansion at z as

$$T_f(z + d) = T_f(z) + Ad + N_z(d). \quad (20)$$

Let $d_k = T_f^k(z) - z$, $p_0 = 0$, $p_{k+1} = r + A p_k$, and $e_k = d_k - p_k$. Then

$$e_{k+1} = A e_k + N_z(d_k). \quad (21)$$

If $\|A\| \leq \gamma$ and $\|N_z(d)\| \leq L \|d\|^2/2$ on the traversed chart, induction gives

$$\|e_m\| \leq \frac{L}{2} \sum_{k=0}^{m-1} \gamma^{m-1-k} \|d_k\|^2. \quad (22)$$

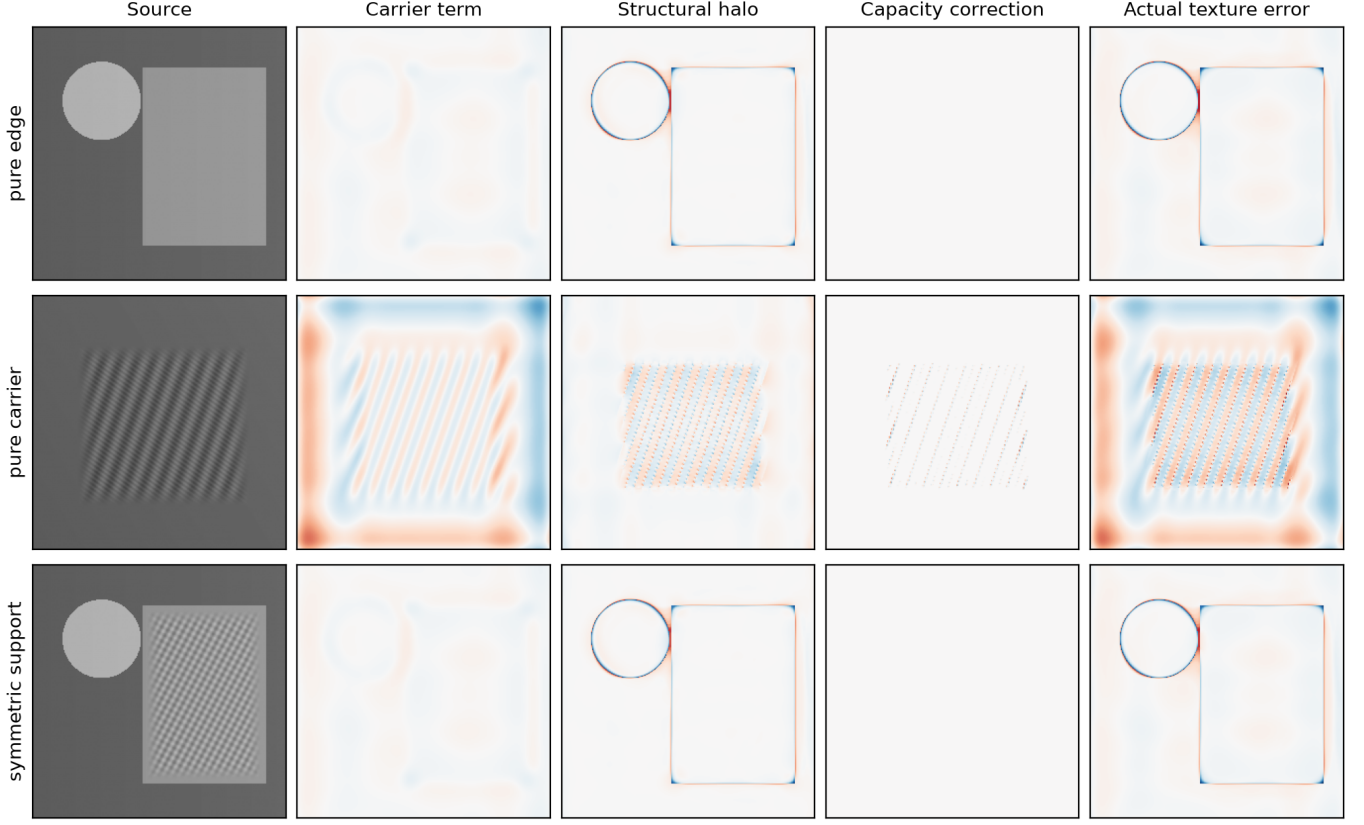


Figure 1: Three representative controls from the six-control scalar-defect archive. The carrier, structural, and capacity columns sum to the actual texture error in the last column, to roundoff. Pure edge and pure carrier isolate the terms; supported texture shows their superposition. Signed error columns share a color scale within each row.

The quadratic assumption is not guaranteed across projection boundaries by semismoothness alone. At $t = (a, 0)$, select the interior derivative and take outward $h = (\epsilon, 0)$. Then $\Pi_a(t + h) - \Pi_a(t) - h = -h$, a first-order remainder. The exterior disk derivative also varies within an unchanged active set. Thus an active set is not an affine chart.

The implemented error additionally contains the depth-two projection term

$$\left\| p_m(A)r - Bp_m(\hat{A})e_1 \right\|. \quad (23)$$

Ordinary passes after each jump refresh the nonlinear disk branches and shorten the chart over which (22) accumulates. Thus the remaining discrepancy from pass 64 consists of finite target tail, Krylov projection, and semismooth chart error. The terms $P_K(u_* - s_1)$ and $-H^K v_*$ of (13) are absent as terms of that particular surrogate identity: this method never forms s_1 , P_K , or a scalar surrogate for z .

5.3 Fixed schedules and the shorter-jump ablation

For prefix p , two tangent products, s ordinary refresh passes, and j jumps, the exact operator-pass cost is

$$C = p + j(1 + 2 + s), \quad (24)$$

where the one counts the exact nonlinear residual evaluation. The public schedules are fixed independently of the source:

	p	horizon	s	j	C
fast	4	18	2	3	19
quality	4	10	2	5	29

We also tested the natural alternative “jump less, refresh once.” A cheap post-jump graph retraction,

$$t \leftarrow Du + \Pi_a(t - Du), \quad (25)$$

restores the dual feasibility $|t - Du| \leq a$ in $O(HW)$ point-wise work and no FFT. Retraction alone prevents instability but loses most of the acceleration. Shorter jumps with one exact refresh won at 128^2 but did not uniformly beat the two-refresh schedules at 256^2 . Reducing only the horizon traded small wins among controls rather than producing dominance. This negative ablation shows that refreshes are nonlinear chart updates, not backtracking overhead. A useful constrained variant is therefore likely to cost about the same total spectral work unless it exploits a stronger bound than (25).

6 C++ realization

The public full-spectral entry point is `bfft_meyer_split_flow_jump`. Its plan owns the ordinary reduced state and

lazily allocates three six-field Krylov vectors plus primal spectra. The implementation performs:

1. a source FFT and ordinary fused prefix;
 2. one exact residual pass without overwriting live spectra;
 3. two streamed tangent passes using (16);
 4. one fixed-chunk five-scalar Gram reduction;
 5. the 2×2 polynomial in registers and one state AXPY;
 6. fixed ordinary refresh passes, repeated by schedule.
- No candidate scan or line search is performed. Texture is emitted as $f - u - w$ and cartoon as $f - v$, guaranteeing exact two-product recomposition.

The Python binding exposes `MeyerPlan.split_flow_jump`; the default full-spectral `split` uses the quality schedule. The effect API and still-image demo expose `quality`, `fast`, `fused64`, and the old hard control. The video demo uses the fixed fast schedule. FACR plans retain their configured ordinary fused recurrence because the current finite-flow tangent is full-spectral; they also fold the finite survivor into effective cartoon.

Native tests compare the C++ output to an independent NumPy implementation of (7)–(19), check exact recomposition, translation equivariance, and bit identity across thread counts, and require both public schedules to beat their equal-cost ordinary controls on the surgical battery.

7 Distance to the objective

A visually accepted finite image is an engineering reference, not a certificate of the discrete constrained model

$$F^* = \min_{u, v \in \mathcal{G}_\mu} J(u) + \frac{\lambda}{2} \|f - u - v\|^2, \quad \mathcal{G}_\mu = \{D^*g : |g(x)| \leq \mu\}. \quad (26)$$

We report $E_{64} = \|v_{\text{out}} - v_{64}\| / \|v_{64}\|$ and late trajectory differences, while separately recovering feasible bounds.

From any retained state define

$$\begin{aligned} p &= \eta_u \Pi_{a_u}(t_u), & q &= D^*p, \\ g &= (\eta_w / c_w) \Pi_{a_w}(t_w), & v_F &= D^*g. \end{aligned} \quad (27)$$

The flux radii are one and μ respectively.

Proposition 7.1 (Feasible objective certificate). *For*

$$\begin{aligned} P &= J(u) + \frac{\lambda}{2} \|f - u - v_F\|^2, \\ L &= \langle f, q \rangle - \frac{\|q\|^2}{2\lambda} - \mu J(q), \end{aligned} \quad (28)$$

one has $L \leq F^* \leq P$.

Proof. The upper bound uses a feasible pair. For the lower bound use $J(u) \geq \langle u, q \rangle$ and $\sup_{v \in \mathcal{G}_\mu} \langle v, q \rangle = \mu J(q)$, then minimize the quadratic. Equivalently, the gap decomposes into nonnegative terms:

$$\begin{aligned} P - L &= J(u) - \langle u, q \rangle + \mu J(q) - \langle v_F, q \rangle \\ &\quad + \frac{\lambda}{2} \|f - u - v_F - q/\lambda\|^2. \end{aligned} \quad (29)$$

□

This needs pointwise projections, local differences, and reductions, with no additional screened solve. It certifies the recovered feasible pair, not automatically the emitted texture. We also record their RMS difference. A small gap does not establish uniqueness of the decomposed images.

8 Engineering evidence

8.1 Archived finite-flow measurements

The August surgical battery uses six designed 256^2 controls and $(\lambda, \mu) = (.05, 40)$ on $[0, 255]$ intensities. Mean texture RMS error to fused 64 is 3.431 for the scalar shortcut, 1.400 for fast versus 1.496 for ordinary 19, and 1.039 for quality versus 1.196 for ordinary 29. Every equal-operator-count comparison improves in that battery.

Archived eight-thread M4 medians on a multiscale crossing are 11.52 ms (fast), 18.16 ms (quality), and 34.60 ms (ordinary 64) at 256^2 . These are different-accuracy operating points. Cold nested Gilles timings in the archive use NumPy/SciPy and different stopping rules; they are historical context, not evidence of a same-implementation speedup.

8.2 September audit and clean timing rerun

The audit adds ordinary 32, 40, and 48 controls; ordinary 2048 and 4096 checkpoints; and the certificate (29). All four comparison sources are resized or retained at 256^2 , including Barbara. Parameters and periodic boundaries are unchanged. The late reference is checked: texture drift from pass 2048 to 4096 is 0.39–1.73%, while the relative recovered objective gap at 4096 is 2.05×10^{-5} to 1.36×10^{-4} . Ordinary 64 differs from pass 4096 by 8.7–22.7% in texture norm.

The original September screen ran under competing CPU load. Its numerical outputs remain valid, but its timings are not used as a clean speed claim. The new record was collected after those workers were stopped. Plan construction and post-run diagnostics are excluded; returned-output allocation is included. Two warmup rounds precede 15 timed rounds, with method order shuffled in each round. Table 1 is generated from serialized measurements.

Equal operator counts do not mean equal time. Zero error to pass 64 is true by definition for ordinary 64, not evidence of objective convergence. The quality schedule has a larger recovered gap than ordinary 32 on all four sources, despite better E_{64} . Its role is finite-state prediction, not direct minimization of that certificate.

8.3 A current-state factor experiment

The audit tests, after the four-pass prefix,

$$z^+ = z + \alpha(T_f(z) - z). \quad (30)$$

This is classical full-state overrelaxation, not a new acceleration principle. Both driving fields are recomputed inside T_f before applying the factor. An experimental native version fuses the state combination and maintains spectra without extra FFTs.

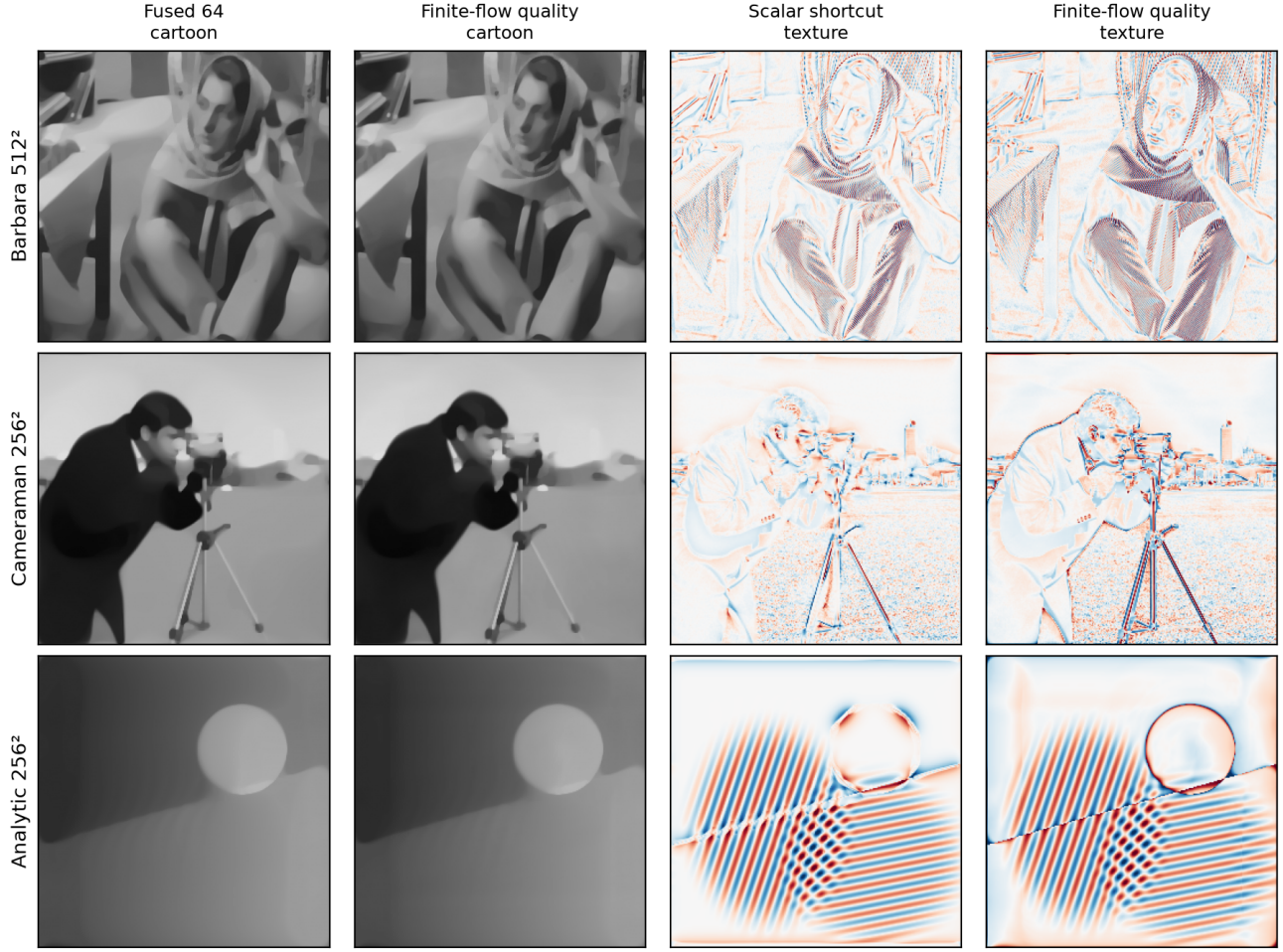


Figure 2: Archived two-product comparison, with survivor folded into effective cartoon. Barbara is 512²; the other two sources are 256². The scalar shortcut retains oscillatory material and exhibits contour error. Finite-flow outputs track the ordinary 64-pass allocation more closely. Texture columns share a color scale within each row.

At $\alpha = 1.75$ and 32 total passes, E_{64} is 59–65% lower than the quality preset on the four audit sources. Recovered gaps are smaller by factors 1.89–3.31. At 64 passes, error to pass 4096 improves by 12–27% against ordinary 64 and the gap improves by 36–54%, but relaxed passes cost more. Same-count gains alone are not matched-time convergence-speed claims.

The factor is not universal. At 128 passes, a screen of eight sources at three parameter pairs leaves all outputs finite, yet $\alpha = 1.75$ improves the gap in only 16 of 19 nontrivial cases. Counterexamples include a ramp, an edge, and a low-frequency carrier. At $\alpha = 1.9$, one ramp has approximately 94.5 times the ordinary gap. Five other cases have ordinary gaps below 10^{-8} and are excluded from gain ratios. Two final ordinary steps improve the gap in every tested finishing-pass ablation, but can worsen agreement with pass 64.

9 Requirements for dynamic factors

A constant multiplier diagnoses available progress; it is not a sufficient method. A production factor should be determined from current transport statistics, carry a stated mathematical

guarantee, and cost materially less than the work it saves. There are three separate requirements:

1. *A valid statistic.* Boundary distance, nonlinear defect, residual, or objective bound must measure the claimed quantity. Active-set persistence alone does not certify a long affine prediction.
2. *A valid decision.* A local slope can suggest a candidate. An established bound or checked inequality must certify its advertised property. A feasible gap comparison certifies that comparison; it does not by itself prove convergence of repeated choices.
3. *A measured cost.* Reductions, stencils, trial projections, and rejected candidates must all be timed. Absence of extra FFTs does not imply negligible overhead.

The certificate (29) enables an initial investigation without changing the model. Dynamic-factor work is maintained as a separate follow-up. No successful adaptive convergence theorem or universal dynamic speedup is asserted in this revision.

Table 1: Clean M4 rerun: eight threads, 256^2 sources, 15 shuffled timed repetitions after two warmup rounds. Each source reports median ms, E_{64} , and recovered objective gap per pixel $G = (P - L)/(HW)$. Barbara is resized to 256^2 .

Method	Barbara			Cameraman			Analytic			Crossing		
	ms	E_{64}	G	ms	E_{64}	G	ms	E_{64}	G	ms	E_{64}	G
Ordinary 32	16.56	0.0628	0.2527	17.43	0.1137	0.3147	17.07	0.1272	0.1639	15.81	0.1349	0.1928
Ordinary 40	21.96	0.0412	0.1675	21.74	0.0817	0.2395	21.71	0.0928	0.1191	20.09	0.1029	0.1618
Ordinary 48	24.97	0.0251	0.1173	26.03	0.0531	0.1913	25.93	0.0601	0.0974	23.52	0.0671	0.1383
Ordinary 64	34.37	0.0000	0.0728	34.96	0.0000	0.1305	34.68	0.0000	0.0711	32.05	0.0000	0.1040
Finite flow fast	11.53	0.0968	0.8921	11.77	0.1582	0.9235	11.72	0.1689	0.7375	10.95	0.1727	0.4824
Finite flow quality	18.03	0.0498	0.4546	18.28	0.0917	0.6279	18.13	0.1111	0.4695	16.91	0.1260	0.3289
Current state 1.75 / 32	18.35	0.0174	0.1371	19.43	0.0374	0.2259	18.40	0.0441	0.1430	17.15	0.0485	0.1736

10 Reproducibility and limitations

The full-spectral implementation is checked against an independent NumPy six-field oracle. Tests cover adjoint signs, tangent finite differences, initialization through trajectory agreement, recomposition, translation equivariance, and thread reproducibility. The audit adds the nonnegative gap decomposition, a zero-gap constant case, a boundary remainder counterexample, and native agreement for experimental factors and refreshes.

Code and evidence reside in `src/detail/meyer_kernel.hpp`, `experiments/meyer_semismooth_state_jump.py`, `experiments/meyer_hard_jump_defect_proof.py`, and `experiments/meyer_transport_audit/`. Raw timing samples, arrays, counterexamples, and reproduction commands are retained. The audit wrapper does not replace production presets.

The study does not establish global convergence of warm one-sweep interleaving for every parameter or boundary condition, image uniqueness, or priority for every ingredient. Finite-flow evidence covers a limited source set and primarily one parameter pair. Factors were selected during exploration on these sources; repeated timing is not held-out validation. The arbitrary-shape route is not included in the finite-flow speed claim.

11 Conclusion

The principal engineering result is a retained transport state that makes the changing coupled problems inexpensive to reconstruct and globally solve. Spectral fusion realizes the same recurrence with less repeated work. Finite-flow approximation and current-state overrelaxation are subsequent choices about how far to advance that state.

A scalar shortcut loses identifiable information; a faithful finite-state jump improves finite-reference agreement; objective certificates reveal progress that a finite image reference cannot measure. Further acceleration should be earned by a cheap, formally stated, continuously updated decision about the current transport, with its full cost included.

Research attribution

Joshuah Rainstar developed the transport-oriented research direction, connected it to earlier superresolution work, and supplied visual and computational falsification criteria. Substan-

tial algebraic exploration, implementation, benchmark scaffolding, and manuscript drafting were performed with AI assistance under that direction. Established model, splitting, and matrix-function ideas are credited in their respective sections. Claims are limited to the stated derivations and executable evidence.

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